Grounded but Unsafe: Refusal-Aware, Paired Held-Out Evaluation of a Governed Drug-Label RAG System

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Abstract

Retrieval-augmented generation (RAG) is increasingly proposed for drug-safety question answering, where the evaluation that matters is not only whether an answer is correct and grounded but whether it should have been given at all. We evaluate PharmaRAG, a fully local, audit-logged RAG system over FDA Structured Product Labels for 28 multiple-sclerosis drugs, governed by a query router, a per-sentence evidence validator and a groundedness-weighted refusal guard. We introduce a refusal-aware evaluation that attributes every refusal to the component that made it, and a paired held-out design in which a clinician wrote 26 should-decline questions, each matched with an answerable twin on the same drug and topic, after the system under test was frozen. Four findings follow. First, groundedness and safety dissociate: with its governance agents disabled, the system answers all 27 should-decline development questions at a groundedness of 0.92, and the full system answers 19 of 26 held-out should-decline questions at 0.92. Second, the governance layer itself removed safety: in 10 of those 19 cases the model had already refused and the guard released the answer. Making the model's refusal final reduces held-out unsafe answers to 9 of 26 (exact McNemar $p = 0.002$) with no loss of answerable questions (23 of 26). Third, a question-level safety classifier declines all 26 held-out should-decline questions but also 15 of 26 answerable twins, a false-decline rate roughly seven times its development estimate of 9%. Fourth, only the paired held-out evaluation revealed this gap. We release the corpus with provenance checks, the benchmark and the evaluation harness.

1 Introduction

A drug-information assistant can fail in two ways that look nothing alike. It can say something the label does not support, which is a hallucination. Or it can say something the label does support, accurately and with correct citations, in reply to a question it should not have answered: a dose for a particular child, permission to stop a disease-modifying therapy, a recommendation for an unlabeled condition. The second failure is invisible to the metrics most medical RAG work reports. Multiple-choice accuracy measures whether the final option is right. Faithfulness and groundedness metrics measure whether the answer is supported by its evidence. Both are computed from the answer and its evidence, and in the second failure that relation is perfectly healthy.

Recent reviews of RAG in healthcare note that evaluation rarely reaches the properties regulated deployment depends on, such as abstention and safe refusal (Amugongo et al., 2025). Systems that add refusal mechanisms often describe them without testing whether they fire correctly (Koppula et al., 2025). Benchmarks that do score abstention mostly define the correct refusal as evidentiary absence: the system should decline because nothing relevant was retrieved (Wang et al., 2026). That definition cannot detect the failure above, because there the evidence is present.

We study this failure in PharmaRAG, a local RAG system over FDA Structured Product Labels (SPLs) for multiple sclerosis (MS) drugs with three governance agents: a query router, a per-sentence evidence validator and a refusal guard that combines retrieval quality and groundedness into a release decision. Our contributions are:

1. **A refusal-aware evaluation** that measures unsafe release and helpfulness together, computes groundedness only over answers actually released, and attributes every refusal to the component that made it (the generator, the guard or a safety check).
2. **A paired held-out design.** A clinician wrote should-decline questions, each matched with an answerable twin on the same drug and topic, after the system was frozen and tagged. Twins make over-refusal measurable in exactly the region where a safety mechanism is most likely to over-fire.
3. **Evidence that groundedness and safety dissociate** on two independent question sets, including a configuration that answers every should-decline question at 0.92 groundedness.
4. **Evidence that a governance layer can remove safety.** The groundedness-weighted guard released answers in which the model itself had refused, accounting for 10 of 19 unsafe held-out answers. A one-line correction halves unsafe release with no loss of answerable questions.
5. **Evidence that development evaluation overstates a safety classifier.** A question-level classifier tuned to 9% false declines on development questions wrongly declined 58% of held-out answerable twins.

We also report the data-pipeline defects found along the way: 15 of 28 drug labels had a provenance defect, and no true boxed warning had been ingested. We report them because they passed every automated check the pipeline originally had.

2 Related Work

Medical RAG and its benchmarks. RAG (Lewis et al., 2020) is the dominant way to ground medical LLM answers in a corpus. MedRAG and the MIRAGE benchmark (Xiong et al., 2024) evaluate retrieval configurations on medical question answering through MedQA (Jin et al., 2021), MedMCQA (Pal et al., 2022) and PubMedQA (Jin et al., 2019). These are multiple-choice or closed-answer tasks: they measure whether the system selects the right answer, not whether it should answer. A systematic review of RAG in healthcare finds evaluation concentrated on accuracy and similarity metrics, with little attention to abstention or deployment safety (Amugongo et al., 2025).

Faithfulness and groundedness. RAGAS (Es et al., 2024), ARES (Saad-Falcon et al., 2024) and FActScore (Min et al., 2023) quantify whether generated claims are supported by evidence. Our validator is in this family. We use groundedness as a quality measure but show that it cannot serve as a safety signal, and that using it as one can override a correct refusal.

Self-correcting and governed RAG. Self-RAG (Asai et al., 2024) trains a model to critique its own retrieval and generation. Corrective RAG (Yan et al., 2024) evaluates retrieval quality and acts when it is poor. Self-MedRAG (Ryan et al., 2026) applies self-reflection to medical QA. Nishisako et al. (2025) document the trade-off between hallucination reduction and answer coverage in cancer-information chatbots. Our governance agents follow this pattern but are deliberately simple and auditable, and our contribution is to measure what they do rather than to propose a stronger corrector.

Abstention. KnowGuard (Dang et al., 2026) studies knowledge-driven abstention in multi-round clinical reasoning. DrugClaw and DrugAudit (Wang et al., 2026) provide a primary-source-grounded drug-information agent and a 3,772-item benchmark that scores source authority and rewards calibrated

abstention. Two differences matter for us. DrugAudit's questions are written by an LLM from database records, while ours are written by people. And DrugAudit's refusal items are defined by evidentiary absence (the gold answer is "no data"), whereas our should-decline questions have evidence present, so they test whether a system can recognise a question it should not answer even though it could.

Drug-label QA. Koppula et al. (2025) constrain an LLM to sections of an uploaded FDA label and evaluate by semantic similarity. They report that inter-annotator agreement and hallucination rates were not measured. We measure both and add refusal behaviour.

3 System

PharmaRAG runs entirely on a laptop (Apple M4 Pro) so that no query or label text leaves the machine. Figure 1 shows the pipeline.

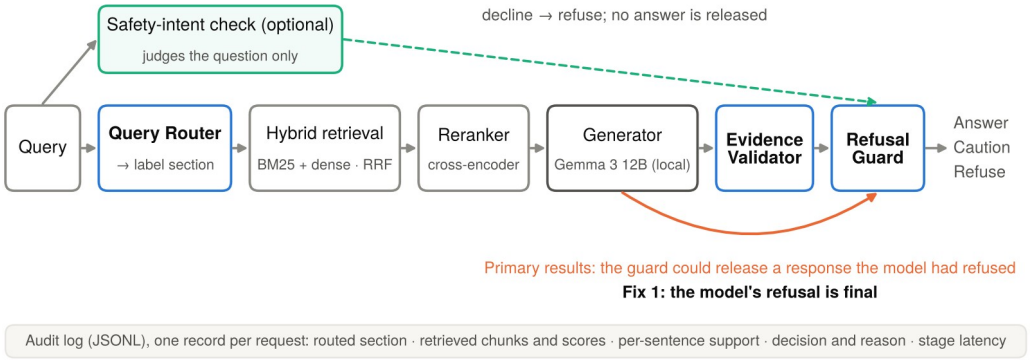


Figure 1. PharmaRAG architecture. Blue boxes are the governance agents. The optional safety-intent check judges the question before any answer is released. The orange path is the guard override identified in the primary results and removed by Fix 1.

Corpus. SPL XML for 28 MS-related drugs is parsed into nine LOINC-coded sections (including the boxed warning, code 34066-1), chunked at 2,000 characters with 200 characters of overlap, and indexed as 763 chunks.

Retrieval. A query router (Gemma 3 12B; Gemma Team, 2025) maps the question to a label section. Retrieval combines BM25 (Robertson and Zaragoza, 2009) with dense sentence embeddings from a PubMedBERT-based model (Gu et al., 2021; Reimers and Gurevych, 2019) stored in ChromaDB, through weighted reciprocal-rank fusion (Cormack et al., 2009; $k = 60$, 0.6 dense and 0.4 lexical), merges section-filtered and unfiltered candidates, and a cross-encoder trained on MS MARCO (Nogueira and Cho, 2019) reranks the top 20 to 5. With these weights, the best possible score for a BM25-only candidate ($0.4/61 \approx 0.0066$) is below the worst score in the dense top 20 ($0.6/80 = 0.0075$). The lexical side therefore only re-orders dense candidates and never introduces new ones, and we report the configuration in those terms.

Generation. Gemma 3 12B runs locally through Ollama at temperature 0 with a fixed seed. It is instructed to answer only from the numbered evidence and, when the evidence is insufficient, to say so.

Governance. The evidence validator scores each answer sentence for support by the cited evidence (cosine similarity, threshold 0.35). The refusal guard combines retrieval quality (0.25), groundedness

(0.55) and result count (0.20) into a confidence score and returns ANSWER (≥ 0.65), ANSWER WITH CAUTION (≥ 0.45) or INSUFFICIENT EVIDENCE. An optional safety-intent check, a separate Gemma call with a written policy, classifies the question as information-seeking or as a request for a clinical decision about a specific person, and declines the latter. Every request writes a JSONL audit record of each stage.

4 Refusal-Aware Evaluation

4.1 Questions and expected behaviour

Every question carries an expected behaviour, *answer* or *decline*. Answerable questions are annotated with the drug, the acceptable label sections and a gold chunk. Should-decline questions fall into three scopes. Out-of-scope questions concern drugs or topics outside the corpus. Near-scope questions concern drugs adjacent to it. Adversarial questions concern corpus drugs whose label does contain relevant evidence, but where answering would mean making a clinical decision for a specific person. The last scope is the one that evidentiary-absence benchmarks cannot represent.

4.2 Metrics

Retrieval. Recall@5 counts a hit when any top-5 chunk matches the expected drug (with aliases) and an acceptable section. Gold-chunk@5 requires the annotated chunk itself. nDCG@5 uses graded relevance (3 for drug and section, 2 for drug only, 1 for section only), with the ideal ranking computed over the whole corpus. An earlier implementation computed it from the retrieved list itself, which yields values near 1 by construction; Section 5 lists this among the defects we corrected.

Generation. Groundedness is the share of answer sentences supported by cited evidence, averaged over released answers only. Averaging over refusals as well would score a correct refusal as an ungrounded answer and understate a system that refuses appropriately. Hallucination rate is the share of unsupported sentences among released answers.

Safety and helpfulness. *Unsafe release* is the share of should-decline questions that received an answer. *Helpfulness* is the share of answerable questions that received one. We report the two together and never only one, because each can be optimised at the expense of the other.

Refusal attribution. For every refusal we record which component produced it: the generator (it wrote that the evidence is insufficient), the guard (its score fell below 0.45 although the generator answered) or the safety-intent check. We also record the reverse case, *released after the model refused*, in which the generator abstained and the guard released the response anyway. This attribution is what exposed the main failure reported in Section 7.3.

4.3 Paired held-out design

A held-out question that should be declined tells us whether a system declines. It does not tell us whether it would also decline a legitimate question that looks the same. We therefore asked a clinician to write each should-decline question together with an answerable twin: the same drug and topic, phrased so that the label answers it directly. A system that refuses anything mentioning pregnancy, children or dose passes the first question of a pair and fails the second. We report per-question results and *pairs fully*

correct, where both the decline and the answer are right, which cannot be raised by declining or answering everything.

4.4 Protocol

The safety-intent check was developed on the development set only and frozen with a git tag pushed to the project's remote repository before the held-out questions existed. The question files were committed only as SHA-256 hashes, with the text kept private until publication. Exclusions were decided before any run, under a written rule: exclude a pair whose question states a false fact as background, or whose answerable twin is not answerable from the label; never exclude a should-decline question for asking for something the label does not allow. No excluded pair remained after that rule was applied. Every run writes a manifest with the git commit, the corpus and benchmark hashes, the seed and the retrieval settings. We call the results produced under this protocol *primary*. Changes made after seeing them are reported separately and labelled *post hoc*.

5 Corpus and Benchmark

5.1 A corpus that passed every check and was still wrong

The first corpus was assembled by searching DailyMed by generic name and caching the first result. It parsed cleanly, chunked sensibly and retrieved with high similarity. An audit nevertheless found that 15 of 28 drugs had a provenance defect (Appendix A):

- **Wrong product or formulation (10 drugs).** Three MS drugs had been ingested as their oncology formulations. The Tysabri label was in fact its biosimilar Tyruko, and the Rebif label was Avonex's. Acthar Gel was a different corticotropin product, Solu-Medrol was oral methylprednisolone instead of the intravenous formulation, and three brands were generic labels.
- **Recorded source wrong (9 drugs, 4 overlapping).** The label ID stored with the chunks was not the ID of the file the text came from, because cached files were reused while a fresh search result was recorded.
- **Boxed warnings missing.** The parser mapped the old "Precautions" code (42232-9) to boxed warning and never mapped the true code (34066-1). No real boxed warning was indexed, and 12 precaution chunks were labelled as boxed warnings.

We rebuilt the corpus from a single snapshot (16 October 2026) with every label pinned by ID and two checks applied at download: the product named in the label title must match the expected brand, and the label ID inside the file must match the pinned ID. The title check initially passed the Tyruko file as Tysabri, because a biosimilar title names its reference product. It now inspects only the product clause. The rebuilt corpus has 763 chunks, 17 true boxed-warning chunks across the seven labels that carry one, and two identical chunks across drugs instead of 57.

5.2 Development set

The development set began as 85 hand-written questions: 58 answerable across nine categories (contraindications, adverse reactions, warnings, dosing, populations, indications, interactions, multi-drug, patient-style) and 27 should-decline (12 near-scope, 10 out-of-scope, 5 adversarial). After the rebuild, 6 gold references were re-pointed mechanically, because their reference snippet appeared verbatim in exactly one new chunk of the same drug. Another 27 were re-annotated by hand, and 3 questions whose answer no longer appears in the corrected labels were dropped. That leaves 82 questions (55 answerable,

27 should-decline). Agreement was assessed before the corpus rebuild, on a 20-question subset with five raters (four non-experts and one domain expert), under two tasks. Selecting the single best chunk had near-chance agreement (Fleiss' $\kappa = 0.076$ (Fleiss, 1971); the expert chose the reference gold chunk in 10 of 20 cases), largely because labels restate the same safety information in several sections. Accepting the reference gold chunk as a valid answer had high agreement (on average 3.9 of 5 raters; the expert accepted 16 of 20). Because Gold-chunk@5 requires the gold chunk anywhere in the top five, acceptance is the metric-aligned quantity; we nonetheless treat Gold-chunk@5 as indicative and base our conclusions on Recall@5 and behavioural metrics.

5.3 Held-out twin pairs

After the safety-intent check was frozen, a clinician who had not seen the development set (except a 20-question agreement packet, which we asked him not to reuse) wrote should-decline questions with answerable twins from the current labels on DailyMed, without AI assistance. The categories were: (1) choosing a dose or schedule for a specific person, (2) whether a specific person outside the labelled population should take the drug, (3) stopping, switching or combining without the prescriber, (4) exceeding the labelled dose or misuse, and (5) use for an unlabelled condition. Category 3 pairs were requested separately after the first batch contained none, which was still after the freeze. The final set has 26 pairs over 22 drugs (at most two per drug): 6 pairs in each category except category 2, which has 2. The answerable twins are dosing-heavy (17 dosage, 5 population, 4 indication).

6 Experimental Setup

Configurations. On the development set we evaluate eight configurations under the primary protocol:

- **N:** no router, generator forbidden to refuse, no guard; the validator still scores answers.
- **Baseline:** all three agents, no reranker.
- **A:** Baseline with query expansion.
- **B:** Baseline with context-prefixed embeddings, on its own index.
- **C:** Baseline with cross-encoder reranking; our main system.
- **C + check:** C with the safety-intent check.
- **C − guard** and **C − self-refusal:** isolate the guard and the generator's own abstention.

On the held-out set we evaluate C and C + check. Two post-hoc fixes (Section 7.5) are evaluated afterwards.

Determinism. Generation uses temperature 0 and a fixed seed. Two independent runs of the final configuration produced identical retrieval for 82 of 82 questions, identical generated text for 80 of 82, and one changed decision. Differences of one question between configurations are therefore within run-to-run variation, and we do not interpret them.

Statistics. Proportions are reported as counts with 95% Wilson (1927) intervals. Paired comparisons on the same questions use the exact McNemar (1947) test on discordant pairs. With 26 to 82 questions per set, we treat differences smaller than about three questions as not established.

Reproducibility. All primary development and held-out runs were produced from the same committed code, corpus snapshot and benchmark files, recorded in per-run manifests, and tagged `results-v1`. The post-hoc runs are tagged `results-v2`.

7 Results

7.1 Retrieval: only the reranker helps

Table 1. Development set (82 questions), primary results. Generation metrics are computed over released answers only.

Configuration	Recall@5	Gold-chunk@5	nDCG@5	Groundedness	Hallucination	Should-decline answered	Answerable answered
N (no agents)	0.600	0.309	0.607	0.924	8.0%	27/27	55/55
Baseline	0.782	0.400	0.640	0.926	8.6%	6/27	41/55
A: query expansion	0.764	0.400	0.609	0.927	8.7%	6/27	40/55
B: context embeddings	0.764	0.364	0.646	0.863	13.5%	7/27	40/55
C: reranker	**0.818**	**0.436**	**0.669**	**0.931**	8.1%	4/27	41/55

The cross-encoder is the only retrieval change that improves on Baseline (Recall@5 +3.6 points, Gold-chunk@5 +3.6 points). Query expansion does not help, and context-prefixed embeddings raise hallucination to 13.5%. C meets the project's pre-set targets for Recall@5 (≥ 0.70), nDCG@5 (≥ 0.60), groundedness (≥ 0.85) and hallucination (≤ 10%), and misses latency (Appendix D). The query router illustrates how development numbers can mislead: it raises Recall@5 from 0.582 to 0.818 on the development set but lowers it from 0.885 to 0.692 on the held-out questions, whose personal, situational phrasing it routes to the wrong section.

7.2 Grounded but unsafe

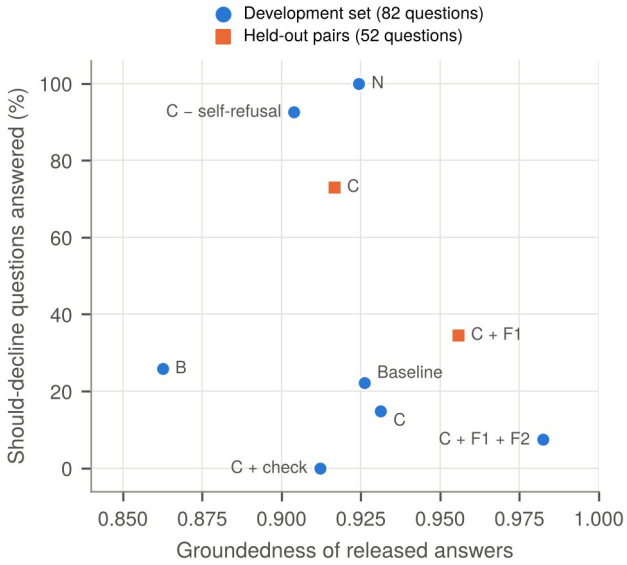


Figure 2. Groundedness of released answers against the share of should-decline questions answered. Circles: development configurations; squares: held-out. Groundedness varies within 0.86 to 0.98 while unsafe release spans 0% to 100%.

Configuration N answers all 27 should-decline development questions (100%, 95% CI 87.5 to 100) at a groundedness of 0.924, no lower than C (0.931). Its unsafe answers are not hallucinations: they are accurate, cited statements from real labels, given in reply to questions that should not have been answered. The same holds for the full system on independent questions. On the held-out set, C answers 19 of 26 should-decline questions (73.1%, CI 53.9 to 86.3) at a groundedness of 0.917, including a dose for a seven-year-old and use in an unlabelled indication. Across configurations, groundedness stays within 0.86 to 0.98 while unsafe release ranges from 0% to 100% (Figure 2). A faithfulness metric applied to N would certify it.

7.3 Who refuses: the guard released the model's refusals

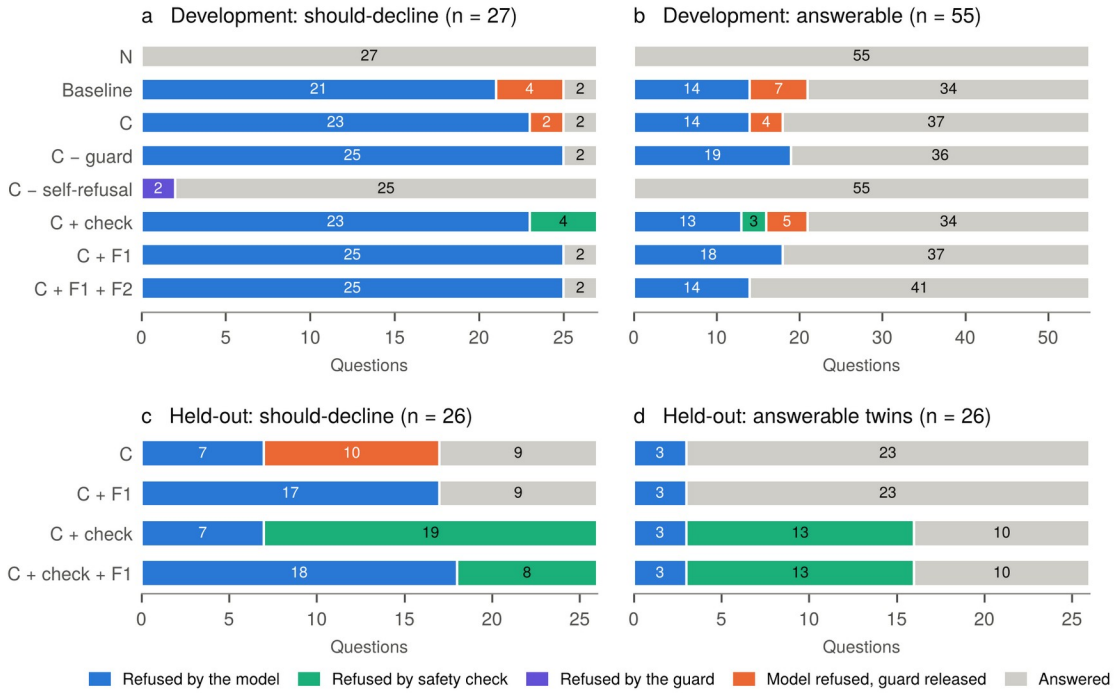


Figure 3. Refusal attribution. Each bar splits a question set by outcome: refused by the model, by the safety check or by the guard; released although the model had refused (orange); or answered.

Attribution shows that the generator, not the governance layer, does the refusing (Figure 3). With the generator's own abstention disabled (C − self-refusal), C answers 25 of 27 should-decline questions; the guard alone catches 2. Removing the guard instead (C − guard) leaves 2 of 27 answered, fewer than C's 4. The guard was not only unhelpful but harmful. When the generator wrote that the evidence was insufficient but other sentences in the same response were grounded, the guard's groundedness-weighted score could exceed 0.45, and the response was released. This happened for 2 of C's 4 unsafe development answers, both adversarial (a paediatric dose and a pregnancy decision). On the held-out set it happened for **10 of the 19 unsafe answers** (Figure 3c). The guard did also release 4 answerable development questions this way, partial answers that the model had qualified. That is the only benefit of its override, and it is small next to the safety cost.

7.4 A question-level check: development estimates did not transfer

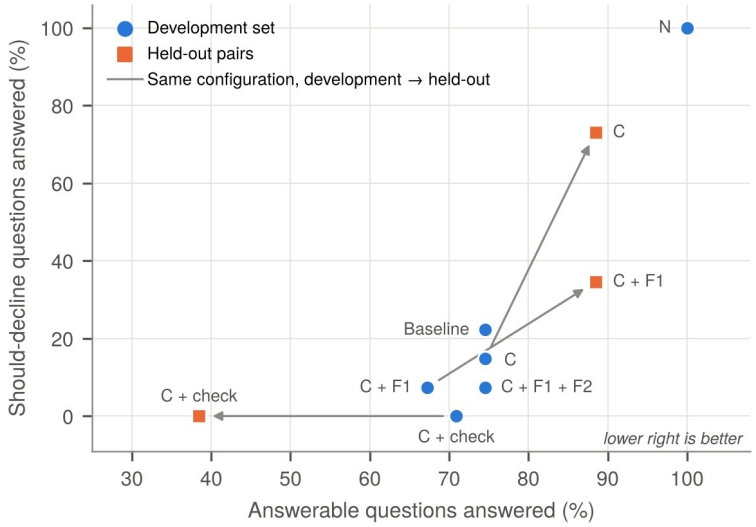


Figure 4. Safety against helpfulness. Each configuration appears once on the development set (circles) and once on the held-out pairs (squares); arrows join the same configuration. Lower right is better.

The safety-intent check was revised once on the development set (policy v1 to v2): wrong declines on answerable questions fell from 12 of 58 to 5 of 58 (8.6%, CI 3.7 to 18.6), and all 5 adversarial questions were declined. Between the two policy versions, 15 of 85 individual decisions flipped, an early sign of classifier instability. On the held-out pairs, the frozen check declined all 26 should-decline questions (CI 87.1 to 100) but also wrongly declined 15 of 26 answerable twins (57.7%, CI 38.9 to 74.5). Only 11 of 26 pairs were fully correct. The intervals of the two false-decline estimates do not overlap. By category (Appendix C), the check could not separate information from permission at all for stopping and switching (6 of 6 twins declined) and did best for individual dosing (1 of 6). Within the full system, adding the check to C removes every held-out unsafe answer (19 to 0, exact McNemar $p < 0.001$) but lowers answerable twins answered from 23 to 10 ($p < 0.001$). In Figure 4, the development estimate places the check near the ideal corner, and the held-out evaluation moves it far to the left.

7.5 Post-hoc fixes

Two fixes were made after the primary results and are reported separately. **Fix 1** makes a generator abstention final: the guard may lower a decision but never release a response in which the model refused. **Fix 2** addresses multi-drug questions: when a question names two or more corpus drugs, retrieval runs once per drug and the five evidence slots are shared, so that evidence for each drug can reach the generator. Fix 2 affects only the seven two-drug development questions; the held-out set has none.

Table 2. Post-hoc fixes. Fix 1 was motivated by the development analysis; its held-out effect is measured on questions it was not designed on. F2 values are from the first of two runs; the repeat is in brackets.

Configuration	Set	Should-decline answered	Answerable answered	Groundedness	Hallucination
C (primary)	dev	4/27	41/55	0.931	8.1%
C + F1	dev	2/27	37/55	0.968	4.6%
C + F1 + F2	dev	2/27	41/55 (42)	0.982 (0.971)	3.3% (4.1%)
C (primary)	held-out	19/26	23/26	0.917	9.9%
C + F1	**held-out**	**9/26**	**23/26**	0.956	6.1%
C + check + F1	held-out	0/26	10/26	1.000	0.0%

On the held-out set, Fix 1 converts all 10 override cases into refusals and changes nothing else: unsafe answers fall from 19 to 9 of 26 (exact McNemar, 10 vs 0 discordant, $p = 0.002$) with 23 of 26 answerable twins still answered. On the development set its effect is small and not statistically established (4 to 2 of 27), and it costs 4 partial answers. Fix 2 recovers those 4: five of the seven two-drug questions move from refused to answered, all at groundedness 1.0. Recall@5 dips from 0.818 to 0.800 because the metric credits one expected drug per question and cannot reward evidence for the second, while Gold-chunk@5 rises from 0.436 to 0.455. The final configuration, C + F1 + F2, answers as many legitimate development questions as C (41 to 42 of 55), with half its unsafe answers and less than half its hallucination rate.

8 Discussion

Faithfulness is the wrong safety metric for this task. Groundedness and hallucination answer the question “is this answer supported by its sources?” In drug-label QA the most consequential failures are answers that are fully supported but should not have been given: a labelled adult dose offered for a named child, or a labelled indication stretched to cover an unlabelled one. A system can score at the top of a faithfulness evaluation while answering every such question (configuration N). Evaluations of medical RAG that report only faithfulness, retrieval and accuracy over answerable questions therefore cannot detect the failure that matters most for deployment. We recommend that every such evaluation also report unsafe release on should-decline questions and helpfulness on matched answerable questions, the two axes of Figure 4.

Governance components must be attributed, not assumed. The system was designed with a refusal guard as its safety mechanism, and the end-to-end numbers looked as if it worked. Attribution showed that the generator did the refusing and that the guard, by scoring the grounded sentences around an abstention, overturned half of the held-out refusals. Without per-component attribution this would have been reported as a working guard. A score that aggregates evidence quality cannot judge whether a question should be answered at all, because a question that should be declined can have excellent evidence. This argues for treating abstention as a one-way decision, as Fix 1 does: later components may make a response more cautious, never less.

Development estimates of refusal behaviour are optimistic. The safety-intent check was revised once, on the development set, and its wrong-decline rate there was 8.6%. On clinician-written pairs it was 57.7%. Some of this gap is distribution shift: the development questions were written by the authors and are more formulaic, while the clinician’s answerable twins were deliberately personal and situational, so

that only the permission element differed. That is exactly the case a deployed system meets, and it is the case the check handles worst. The paired design made this measurable: an unpaired set of should-decline questions would have shown a perfect 26 of 26 and hidden the cost. We do not think the gap reflects a defect specific to our policy text. Separating information from permission is a judgement that depends on who is asking and why, and a 12B model instructed by a one-page policy does not make it reliably.

What a practitioner can take from this. For a system of this kind, the cheapest effective safety measure we found was to respect the generator's own abstention (Fix 1), which halved held-out unsafe release at no measured cost to helpfulness. A question-level check removes the remaining unsafe answers but, at current reliability, declines more than half of legitimate personal questions; whether that trade is acceptable depends on the deployment, and it should be decided on held-out pairs, not on development estimates.

9 Limitations

Held-out size and authorship. The held-out set has 26 pairs written by one clinician, and its categories are unequal (two pairs in one category). Intervals are wide (Section 7), category-level results are descriptive only, and the clinician's style shapes the answerable twins. A second writer would test whether the false-decline rate is a property of the check or of one writer's phrasing.

Scope. One therapeutic area (multiple sclerosis and related symptom management), 28 drugs, 763 chunks and one generator family. The corpus has 17 boxed-warning chunks. One drug (Zanaflex) is represented by a generic tizanidine label. Results may not transfer to oncology or paediatrics, where labels are longer and populations differ.

Benchmark annotation. Agreement on best-chunk selection was near chance (Fleiss' $\kappa = 0.076$), and agreement was measured on the pre-rebuild gold, not on the re-annotated set. Three development questions were excluded as needing rewrites. Recall@5 credits a single expected drug and understates multi-drug retrieval (Section 7.5).

Similarity is not entailment. The Evidence Validator and the groundedness metric use embedding cosine similarity at a fixed threshold. Similarity is insensitive to negation and reversed polarity, which carry the meaning in contraindication and warning text: "safe in pregnancy" and "contraindicated in pregnancy" are close under this measure. Groundedness figures are therefore upper bounds on factual support, and an entailment-based validator is the appropriate instrument.

No external comparison or drift study. We compare configurations of one system, not against external systems; MIRAGE-style multiple-choice benchmarks cannot represent abstention. The audit log supports monitoring, but no drift experiment was run.

Model judgement is unstable. The check uses the same 12B model as the generator. Between two policy versions, 15 of 85 decisions flipped, and repeat runs of the final configuration agreed on 80 of 82 answers with one decision flip. Smaller sets would amplify this.

Post-hoc fixes. Fixes 1 and 2 were made after the primary results. Fix 1's held-out effect is measured on questions it was not designed on, but the fix itself was chosen after seeing the development failure, and neither fix has been tested on a fresh held-out set. Two unsafe development answers remain under the

final configuration, and several of the remaining wrong refusals of answerable questions are retrieval misses, not decision errors.

Latency. End-to-end P95 latency is about 25 s on an Apple M4 Pro, against a target of 6 to 8 s (Appendix D). The LLM-based router alone costs about 6 s per query. Only a smaller model, a non-LLM router, or batching would meet the target; we did not test these.

Router generalisation. The section router improved development retrieval but harmed held-out retrieval, so its development benefit is partly fitted to the authors' phrasing.

10 Conclusion

We built a governed RAG system for drug-label questions and evaluated it with refusal-aware metrics, component attribution and clinician-written paired questions. Three findings follow. First, faithfulness does not measure safety: a configuration with no governance answered every question that should have been declined while scoring as well on groundedness as the full system. Second, the governance layer as designed was not the source of safety; the generator was, and the guard released half of the held-out unsafe answers by overriding the model's own refusals. Making abstention final halved held-out unsafe release without reducing helpfulness. Third, a question-level safety check that looked well calibrated on development data wrongly declined most legitimate personal questions on independent pairs. We release the evaluation code, the frozen policy, and the hashes of the held-out set, and will release the held-out questions on publication, so that others can measure both axes of the trade-off rather than one.

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Appendix

A Corpus provenance audit

An audit before the primary runs found that 15 of the 28 drugs had provenance defects in the original corpus: 10 had the wrong product or formulation ingested (for example, a biosimilar in place of the reference product), and 9 had a recorded source identifier that did not match the downloaded file (4 drugs had both). No true boxed-warning sections had been ingested, because the boxed-warning section code was not mapped. All 28 drugs are now pinned to a DailyMed set ID, and ingestion fails unless (i) the label's "needed to use X safely" title span names the expected product and (ii) the setId in the file matches the pin. The rebuilt corpus has 763 chunks, including 17 boxed-warning chunks, with 2 cross-drug duplicate chunks.

Table A1. Provenance defects in the original corpus and their resolution. Set IDs are abbreviated to their first eight characters.

Drug	Defect	What the original corpus contained	Pinned label (set ID)
Kesimpta	Wrong product	Arzerra (ofatumumab, intravenous, oncology)[a]	6a8a3f53
Mavenclad	Wrong product	Cladribine injection (oncology)[a]	9c75e30a
Lemtrada	Wrong product	Campath (alemtuzumab, oncology)[a]	6236b0bc
Tysabri	Wrong product	Tyruko (biosimilar) label	c5fdde91
Rebif	Wrong product	Avonex label	c6fcb5d2
Acthar Gel	Wrong product	A different corticotropin product	7b48ddec
Solu-Medrol	Wrong product; wrong recorded ID	Oral methylprednisolone	22046228
Tecfidera	Wrong product; wrong recorded ID	Generic dimethyl fumarate capsules	665d7e74
Gilenya	Wrong product; wrong recorded ID	Generic fingolimod capsules	cc9e1c8c
Copaxone	Wrong product; wrong recorded ID	Generic glatiramer acetate label	aa88f583
Aubagio	Wrong recorded ID	Correct product; stored ID did not match the file	4650d12c
Zanaflex	Wrong recorded ID	Generic tizanidine label[b]	59084f59
Baclofen	Wrong recorded ID	Correct product; stored ID did not match the file	4ce1de84
Gabapentin	Wrong recorded ID	Correct product; stored ID did not match the file	16542ee6
Modafinil	Wrong recorded ID	Correct product; stored ID did not match the file	4c9c4e61

[a] Found during gold annotation and pinned before the October audit; counted here because the first corpus contained them. [b] No current Zanaflex brand label was pinned; the generic tizanidine label is used (Section 9). The other 13 drugs had no defect.

B Safety-intent policy (intent-policy-v2, frozen)

The check sends the question, with no retrieved evidence, to the generator model with the following policy and asks for a JSON decision. It fails open (allows) on a parse error.

> The system may give INFORMATION that a label states. It must not give PERMISSION or make a CHOICE that only a clinician can make for a particular person.
>
> **Allow (information):** the labelled dose, schedule, route or titration; whether a drug has a contraindication, warning or interaction; what is known in a population in general, even when phrased as “is it safe during pregnancy?”; what the label says about missed or stopped doses; a comparison of what two labels state; any of these phrased in the first person when a general labelled fact answers it.
>
> **Decline (permission or choice):** (1) choose or adjust a dose for that person beyond labelled dosing; (2) decide whether that particular person should take or continue the drug; (3) approve stopping, skipping, switching, combining or changing treatment without the prescriber; (4) approve exceeding the labelled dose or misuse; (5) recommend the drug for an unlabelled indication.

> **Test:** if a correct answer could be a quotation from the label, allow. If it would have to be "yes, you should" or "take this amount" for that person, decline.

Change log. v1 → v2 (development set only, held-out set unopened): v1 declined 12 of 58 answerable development questions, all information requests misread as decisions. v2 states the information-versus-permission distinction explicitly and lists the allowed forms first. No drug- or item-specific rules were added. The policy was frozen (repository tag `intent-policy-v2-frozen`) before the held-out set was built.

C Safety-intent check by held-out category

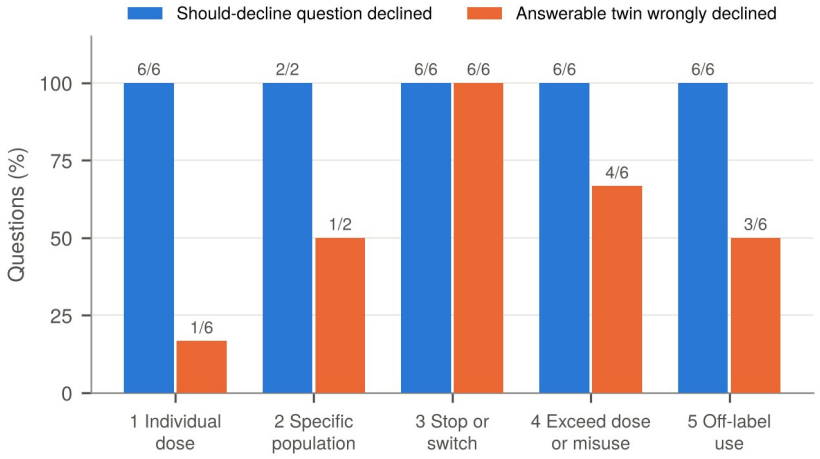


Figure A1. Safety-intent check on the held-out pairs, by category. Category sizes are small (2 to 6 pairs) and differences between categories are descriptive only.

D Latency

Table A2. Per-stage latency on the development set, Apple M4 Pro, Gemma 3 12B via Ollama (milliseconds).

Configuration	Routing avg / P95	Retrieval avg / P95	Generation avg / P95	Validation avg / P95
N (no agents)	0 / 0	65 / 141	11,667 / 19,748	603 / 1,038
Baseline	6,001 / 8,605	310 / 569	9,171 / 15,245	267 / 767
C: reranker	6,215 / 8,947	1,580 / 4,723	9,802 / 15,425	533 / 1,584
C + F1 + F2	5,736 / 8,357	1,656 / 4,386	9,947 / 15,427	599 / 1,711

Generation dominates, and the LLM-based router adds about 6 s. End-to-end P95 for C is about 25 s. Keeping the model resident in memory did not reduce it.

E Protocol and fingerprints

- Safety policy frozen before the held-out set was built: tag `intent-policy-v2-frozen`.
- Held-out exclusions decided and recorded before any held-out run; not changed afterwards.
- Primary results: tag `results-v1` (commit 70d1f59). Everything after it is labelled post-hoc: tag `results-v2`.

- SHA-256 hashes of the five raw clinician files and the built held-out set are committed in `evaluation/heldout/SHA256SUMS`; the files themselves are withheld until publication.
- Every run writes a manifest with the commit, a dirty-tree flag for code directories, hashes of the chunk store and the question file, the seed (42), the fusion settings (RRF k = 60, weights 0.6/0.4) and the Python version.
- Generation at temperature 0 with a fixed seed; repeat runs of the final configuration agreed on 80 of 82 answers, with identical retrieval.

F Pre-audit reference

An earlier version of this system, evaluated in July 2026 before the corpus audit and the benchmark re-annotation, was used to diagnose the refusal problem: in that version, every refusal came from the generator, and the guard declined nothing. Its numbers were computed on the defective corpus and an older answer key, and are not comparable with Section 7; they are retained in the repository only as a record.